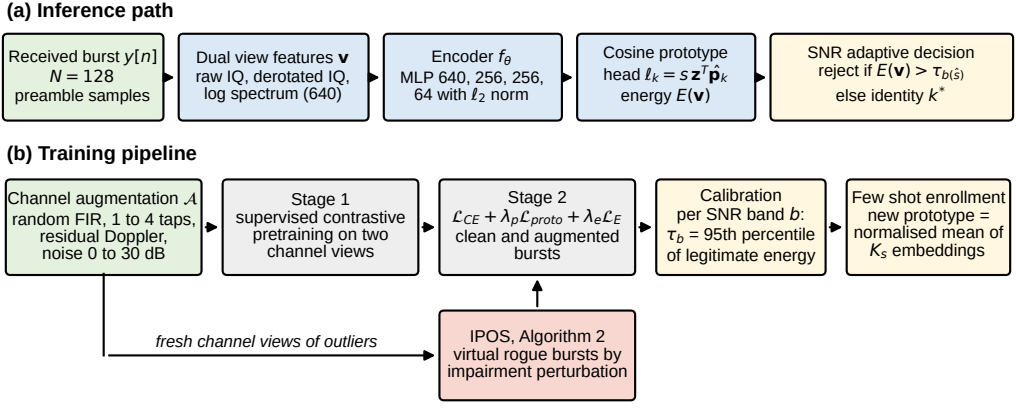
(a) Inference path
Received burst $y[n]$ $N = 128$ preamble samples
Dual view features $\mathbf{v}$ raw IQ, derotated IQ, log spectrum (640)
Encoder f_θ MLP 640, 256, 256, 64 with ℓ_2 norm
Cosine prototype head $\ell_k = s\,\mathbf{z}^T\hat{\mathbf{p}}_k$ energy $E(\mathbf{v})$
SNR adaptive decision reject if $E(\mathbf{v}) > \tau_{b(\hat{s})}$ else identity k^*
(b) Training pipeline
Channel augmentation $\mathcal{A}$ random FIR, 1 to 4 taps, residual Doppler, noise 0 to 30 dB
Stage 1 supervised contrastive pretraining on two channel views
Stage 2 $\mathcal{L}_{CE} + \lambda_p\mathcal{L}_{proto} + \lambda_e\mathcal{L}_E$ clean and augmented bursts
Calibration per SNR band b: τ_b = 95th percentile of legitimate energy
Few shot enrollment new prototype = normalised mean of K_s embeddings
fresh channel views of outliers
IPOS, Algorithm 2 virtual rogue bursts by impairment perturbation